

RANOpt AI

AI-Assisted 5G/6G RAN Optimization Platform

Executive overview of the platform.

Acronyms

RAN: Radio Access Network – wireless network connecting users.

AI: Artificial Intelligence – software that learns to make better decisions.

RRM: Radio Resource Management – shares radio resources efficiently.

MAC: Medium Access Control – decides who transmits next.

Industry Challenge

Why AI is needed in modern RANs

Increased network complexity
More spectrum and antennas
AI optimizes automatically.

Acronyms

AI: Artificial Intelligence – software that learns to make better decisions.
RRM: Radio Resource Management – shares radio resources efficiently.
MAC: Medium Access Control – decides who transmits next.
CSI: Channel State Information – report describing radio channel quality.

Platform Vision

Goals of RANOpt AI

Platform optimized for 5G Advanced and emerging 6G eMBB networks.

Low-band FDD for wide-area coverage.

FDD + TDD + Carrier Aggregation for high-capacity eMBB.

Current product focuses on eMBB (not IoT/eMTC/URLLC).

Acronyms

RRM: Radio Resource Management – shares radio resources efficiently.

MAC: Medium Access Control – decides who transmits next.

CSI: Channel State Information – report describing radio channel quality.

PPO: Proximal Policy Optimization – RL algorithm improving decisions gradually.

Platform Architecture

End-to-end system flow

User Interface → Simulator → AI Engine → RRM → MAC
Scheduler → KPI Dashboard
Supports Low-band FDD.
Supports FDD + TDD Carrier Aggregation.
3GPP CDL channel models.

Acronyms

MAC: Medium Access Control – decides who transmits next.

CSI: Channel State Information – report describing radio channel quality.

PPO: Proximal Policy Optimization – RL algorithm improving decisions gradually.

DQN: Deep Q-Network – neural-network reinforcement learning.

Simulation Engine

3GPP-aligned modelling

Supports FDD and FDD+TDD deployment.
Massive MIMO.
3GPP CDL channel models.
Carrier Aggregation for eMBB.

Acronyms

CSI: Channel State Information – report describing radio channel quality.
PPO: Proximal Policy Optimization – RL algorithm improving decisions gradually.
DQN: Deep Q-Network – neural-network reinforcement learning.
UE: User Equipment – phone or modem.

AI Engine

Current and future AI

Current rule-based AI
Future PPO and DQN RL

Acronyms

PPO: Proximal Policy Optimization – RL algorithm improving decisions gradually.

DQN: Deep Q-Network – neural-network reinforcement learning.

UE: User Equipment – phone or modem.

gNB: 5G base station.

AI-Optimized RRM

Resource optimization

Scheduler optimization
PRB allocation
Power and load balancing

Acronyms

DQN: Deep Q-Network – neural-network reinforcement learning.

UE: User Equipment – phone or modem.

gNB: 5G base station.

PRB: Physical Resource Block – smallest radio resource unit.

MAC Scheduler

Scheduling intelligence

PF, Max CQI, Round Robin
Future AI scheduler

Acronyms

UE: User Equipment – phone or modem.

gNB: 5G base station.

PRB: Physical Resource Block – smallest radio resource unit.

MCS: Modulation and Coding Scheme – selects transmission speed/reliability.

AI-Assisted CSI

Reducing uplink signalling

AI predicts CSI from previous observations.
Reduce UL CSI signalling overhead.
Increase scheduler decision accuracy.
Improve DL spectral efficiency.
Maintain throughput with fewer CSI reports.

Acronyms

gNB: 5G base station.

PRB: Physical Resource Block – smallest radio resource unit.

MCS: Modulation and Coding Scheme – selects transmission speed/reliability.

KPI: Key Performance Indicator – measurable network performance.

RL Framework

Learning by interaction

1. Observe – AI monitors CSI, active UEs, PRB utilization, cell loading, SINR/CQI and traffic demand.
 2. Decide – PPO selects the best scheduler, PRB allocation, carrier (FDD/FDD+TDD), MCS and beam configuration.
 3. Execute – The simulator applies the selected RRM action through the MAC scheduler and recalculates network KPIs.
 4. Measure Reward – The simulator evaluates cell throughput, edge throughput, spectral efficiency, fairness and reduced UL CSI signalling.
 5. Learn – PPO updates its policy from the measured reward so future decisions become progressively better.
- Decision Loop: Observe → Decide → Execute → Measure
Reward → Learn → Repeat.
- Training is performed offline over many simulation episodes; the web interface uses the trained model for real-time inference.

Acronyms

RL: Reinforcement Learning – AI learns from trial and reward.

PPO: Proximal Policy Optimization – RL algorithm that improves decisions gradually.

RRM: Radio Resource Management – controls allocation of radio resources.

MAC: Medium Access Control – schedules transmissions.

KPI: Key Performance Indicator – measures network performance.

Demo Dashboard

KPIs and visualization

Interactive engineering dashboard for eMBB evaluation

Coverage Map – predicts service area and cell-edge coverage.

Capacity Map – shows throughput distribution using FDD/FDD+TDD and CA.

AI On/Off – compare identical scenarios instantly.

Scheduler – Round Robin, PF, Max CQI or AI.

KPI Dashboard – Throughput, Spectral Efficiency, Fairness and CSI reduction.

Acronyms

MCS: Modulation and Coding Scheme – selects transmission speed/reliability.

KPI: Key Performance Indicator – measurable network performance.

CA: Carrier Aggregation – combines carriers for higher speed.

CDL: Clustered Delay Line – realistic 3GPP channel model.

Performance Gains

Illustrative improvements

Illustrative AI Performance Improvements

Cell Throughput ↑ – more traffic carried per cell.

UE Throughput ↑ – higher average user data rates.

Edge Throughput ↑ – better experience at cell boundary.

Spectral Efficiency ↑ – more bits/s/Hz from the same spectrum.

Fairness ↑ – more balanced resource sharing.

UL CSI Signalling ↓ – fewer UE feedback reports with AI prediction.

Acronyms

KPI: Key Performance Indicator – measurable network performance.

CA: Carrier Aggregation – combines carriers for higher speed.

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3GPP Alignment

Release roadmap

Rel-15 NR

Rel-16 CA

Rel-18 AI

Rel-19+ AI-native

Acronyms

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CDL: Clustered Delay Line – realistic 3GPP channel model.

Future Roadmap

Evolution to AI-native 6G

V14.6 -available now)
V15 -under development)
V16 -Future Digital Twin

Acronyms

CDL: Clustered Delay Line – realistic 3GPP channel model- with 3D Beam Forming

Summary

Key takeaways

Commercial-ready eMBB optimization platform.

Supports FDD and FDD+TDD.

3GPP aligned.

AI-native evolution toward 6G.